A decimation program produces 28 output samples from an input of 96 samples with factor and Identify the coding error.

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INTRODUCTION

- Decimation is the process of reducing the sampling rate of a digital signal by keeping every M-th sample and discarding the remaining samples.
- The decimation factor (M) determines how much the sampling rate is reduced.
- The expected number of output samples is calculated as:
- $\text{Output Samples} = \dfrac{\text{Input Samples}}{\text{Decimation Factor}}$
- In a correct decimation program, the output sample count must match this expected value.
- If the actual output differs from the expected output, it indicates a coding or implementation error.

PROBLEM SOLVING

given:

- Input samples = 96
- Decimation factor = 4
- Program output = 28 samples

Calculate the expected output:

$$\text{Expected Output} = \frac{96}{4} = 24 \text{ samples}$$

- Compare it with the actual output (28 samples).
- Conclude that 28 is incorrect.

```
clc;
clear;
close all;
% Given data
input_samples = 96;
decimation_factor = 4;
actual_output = 28;
% Calculate expected output
expected_output = input_samples / decimation_factor;
% Display results
fprintf('Input Samples      = %d\n', input_samples);
fprintf('Decimation Factor  = %d\n', decimation_factor);
fprintf('Expected Output    = %d\n', expected_output);
fprintf('Actual Output      = %d\n', actual_output);
% Check for coding error
if actual_output == expected_output
    fprintf('\nProgram is CORRECT.\n');
else
    fprintf('\nProgram is INCORRECT.\n');
    fprintf('Coding Error Detected!\n');
    fprintf('Expected %d output samples but obtained %d.\n', ...
            expected_output, actual_output);
    fprintf('Possible reasons:\n');
    fprintf('1. Wrong loop limit.\n');
    fprintf('2. Incorrect indexing.\n');
    fprintf('3. Off-by-one error.\n');
    fprintf('4. Incorrect sample counter increment.\n');
end
```

OUTPUT

untitled5.mlx untitled6.mlx untitled.mlx untitled * untitled2 * untitled3 *

```
fprintf('Decimation Factor  = %d\n', decim
fprintf('Expected Output    = %d\n', expec
fprintf('Actual Output      = %d\n', actua

% Check for coding error
if actual_output == expected_output
    fprintf('\nProgram is CORRECT.\n');
else
    fprintf('\nProgram is INCORRECT.\n');
    fprintf('Coding Error Detected!\n');
    fprintf('Expected %d output samples bu
        expected_output, actual_output);
    fprintf('Possible reasons:\n');
    fprintf('1. Wrong loop limit.\n');
    fprintf('2. Incorrect indexing.\n');
    fprintf('3. Off-by-one error.\n');
    fprintf('4. Incorrect sample counter i
end
```

```
Input Samples      = 96
Decimation Factor  = 4
Expected Output    = 24
Actual Output      = 28

Program is INCORRECT.
Coding Error Detected!
ted 24 output samples but obtained 28.
Possible reasons:
1. Wrong loop limit.
2. Incorrect indexing.
3. Off-by-one error.
4. Incorrect sample counter increment.
```

THANK YOU